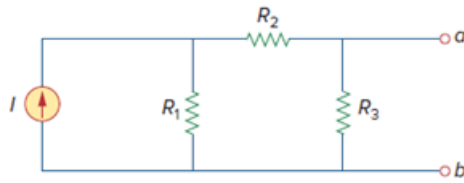


Problem

Using Fig. 4.113, design a problem to help other students better understand Norton equivalent circuits.

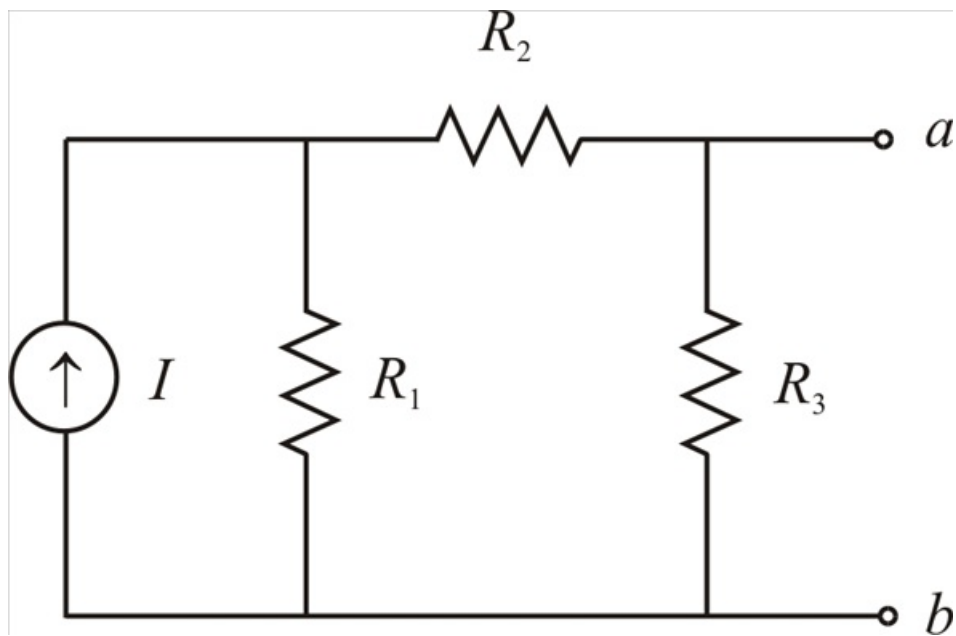
Figure. 4.113:



Step-by-step solution

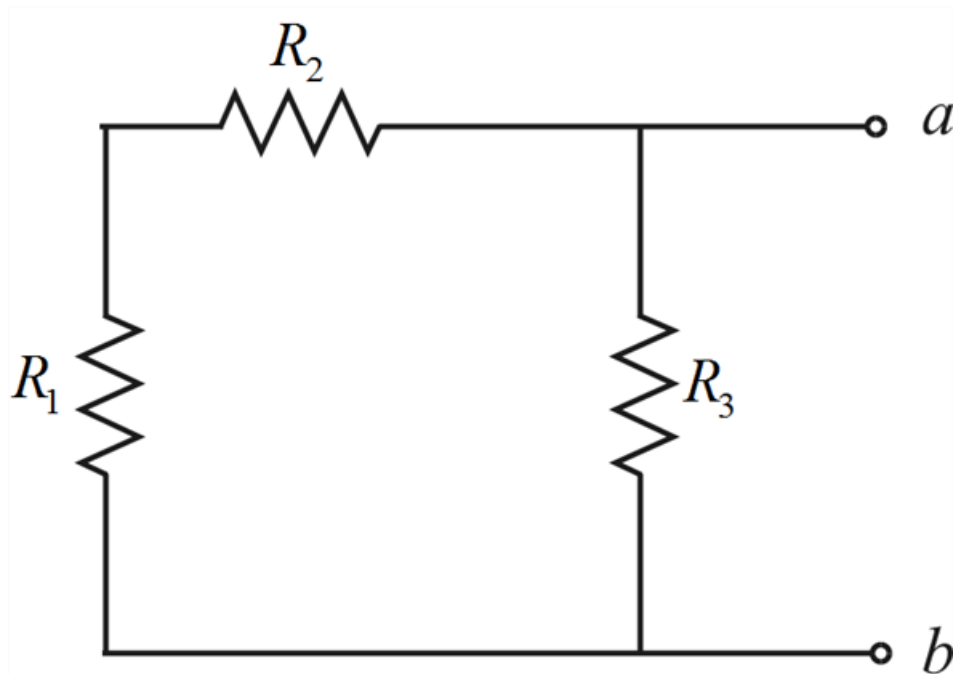
Step 1 of 7

Consider the circuit



Step 2 of 7

To find Norton resistance open circuit the current source



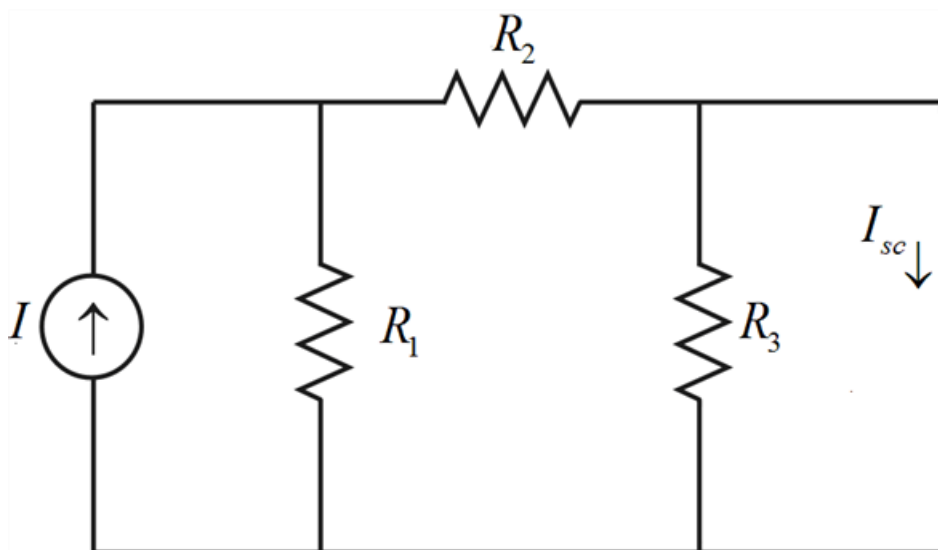
Step 3 of 7

$$R_N = R_3 \parallel (R_2 + R_1)$$

$$R_N = \frac{R_3 (R_2 + R_1)}{R_1 + R_2 + R_3}$$

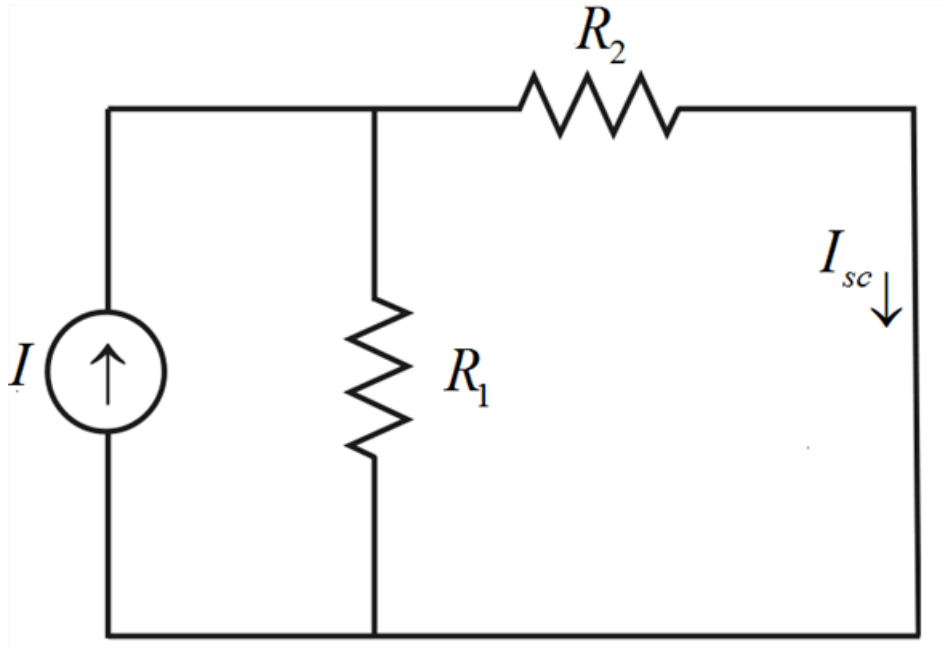
Step 4 of 7

In order to determine I_N we short circuit $a-b$



Step 5 of 7

Short circuit is in parallel with R_3 so R_3 can be removed from the circuit



Step 6 of 7

By current division method

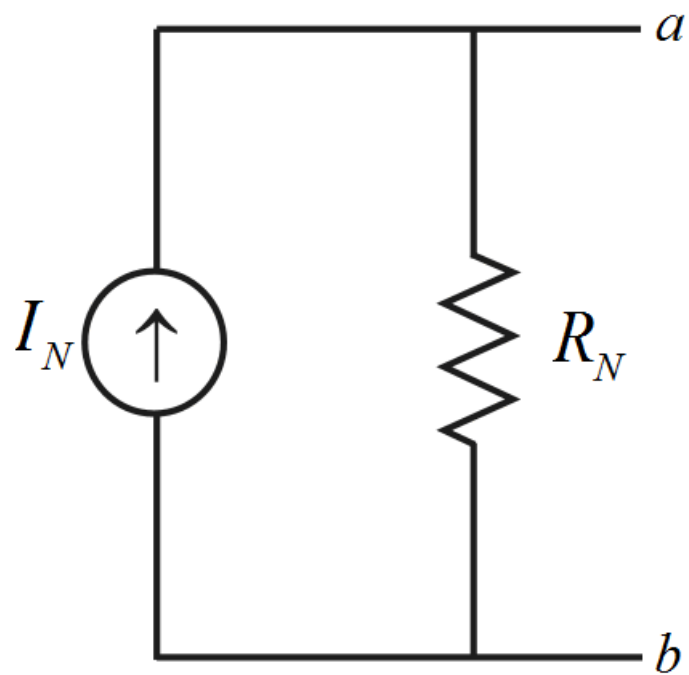
$$I_{sc} = I \frac{R_1}{R_1 + R_2}$$

$$I_{sc} = I_N$$

$$I_N = I \frac{R_1}{R_1 + R_2}$$

Step 7 of 7

Norton equivalent circuit is



Where

$$I_N = I \frac{R_1}{R_1 + R_2}$$

$$R_N = \frac{R_3 (R_2 + R_1)}{R_1 + R_2 + R_3}$$